

Intelligent System for Predictive Diagnostics of Power Transformers

based on DGA: Development of an
ML-pipeline using gradient boosting



Problem & Relevance: Business Pain Points

Critical Infrastructure

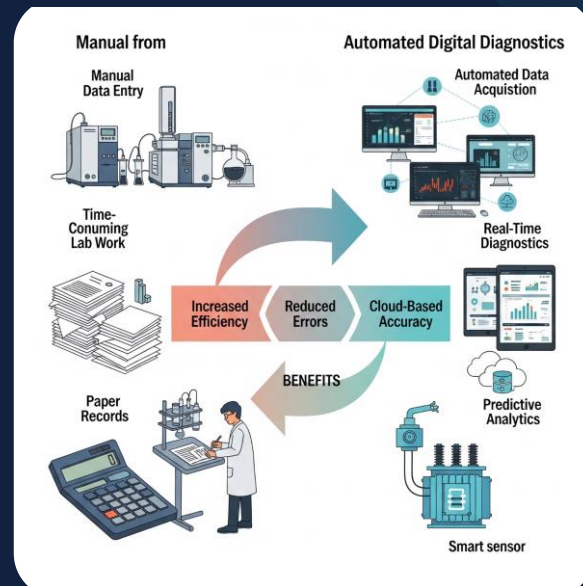
Power transformers are the key and most expensive components of the power grid.

Inefficient Maintenance Regulations

Traditional maintenance leads to accidents or excessive costs.

Human Factor

Engineers manually process giant flows of DGA data.



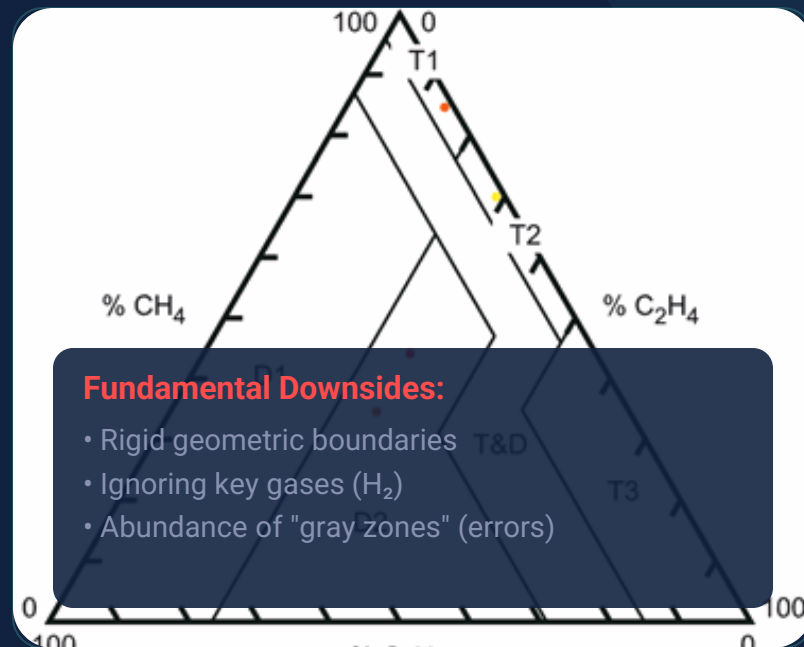
GOAL: EARLY WARNING SYSTEM (ACCURACY > 90%)

Traditional Engineering Methods (Baseline)

Established Standards

Graphical and tabular diagnostic methods have been used in the power industry for decades:

- Duval Triangle
- Key Gas Method
- Rogers Ratio Method



Mathematical Imperfection of Classical Methods

Classification Results (Baseline)

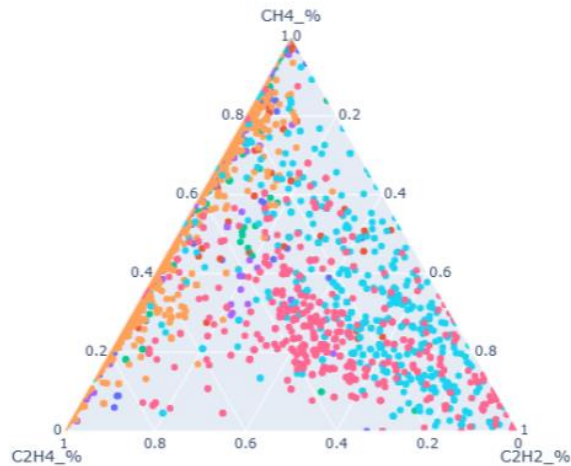
~47%

OVERALL ACCURACY

1%

RECALL FOR NORMAL (NF)

- **Critical blindness:** Key gas method fails to see fault details.
- **False alarms:** Duval classifies healthy objects as defective.
- **Conclusion:** Physical rules from the last century are insufficient for real-world data.



VISUALIZATION: "MESS" IN THE TRIANGLE CENTER

ML Solution Architecture: Engineering Pipeline

Physics and Data Science Integration

Instead of manual analysis, we combined industry expertise and mathematical methods within a unified scikit-learn OOP pipeline.

Custom Transformers

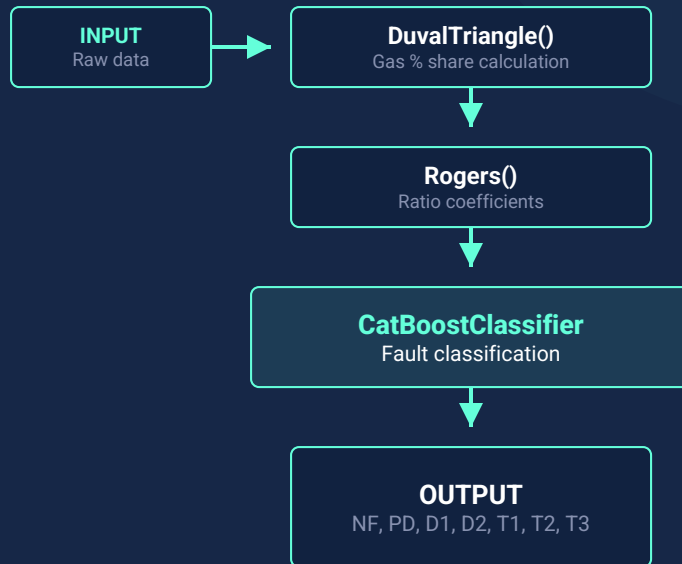
BaseEstimator +
TransformerMixin

Feature Engineering

Feature generation (Duval,
Rogers)

Full Automation and Protection

The model is protected from Data Leakage and is fully ready for deployment on real assets.



Model Training and Optimization

Algorithm: CatBoostClassifier

Ideal for non-linear relationships in physicochemical data tables.

Imbalance Protection

auto_class_weights='Balanced' — equal sensitivity to frequent overheating and rare discharges.

Validation and Search

GridSearchCV using 5-fold stratified cross-validation (StratifiedKFold).

GridSearchCV Hyperparameters:

model__iterations: [100, 200, 300]

model__learning_rate: [0.01, 0.1, 0.2]

model__depth: [3, 4, 5]

model__l2_leaf_reg: [1, 3, 5]

duval_method__param: [True, False]

rogers_method__param: [True, False]

Total combinations: **324**
(across 5 folds = 1620 model fits)

Modeling Results: AI Triumph

Key Performance Indicators

92%

ACCURACY

0.874

MACRO F1-SCORE

- **Explosive Growth:** Accuracy increased from 47% to 92% (2x increase).
- **Normal (NF) Restored:** Precision 99%, Recall 96% — no more false alarms.
- **Complex Defects:** T2 and D1 are now recognized with F1 > 80% (was 0%).

Classification Report (Final)

Class	Precision	Recall	F1-score
NF (Normal)	0.99	0.96	0.97
D1 (Discharges)	0.82	0.85	0.83
T2 (Overheating)	0.88	0.81	0.84
Other	0.90	0.92	0.91
Macro Avg	0.89	0.86	0.8742

Confusion Matrix of the Final CatBoost Model

Distribution Accuracy Analysis

The Confusion Matrix demonstrates the high selectivity of the model between defect classes.

- **Diagonal Dominance:** Most predictions are concentrated on the main diagonal, confirming an overall accuracy of 92%.
- **Minimal Errors:** The model practically never confuses discharges (D1/D2) with thermal defects (T1-T3).
- **NF Class:** Perfect separation between normal and emergency states.

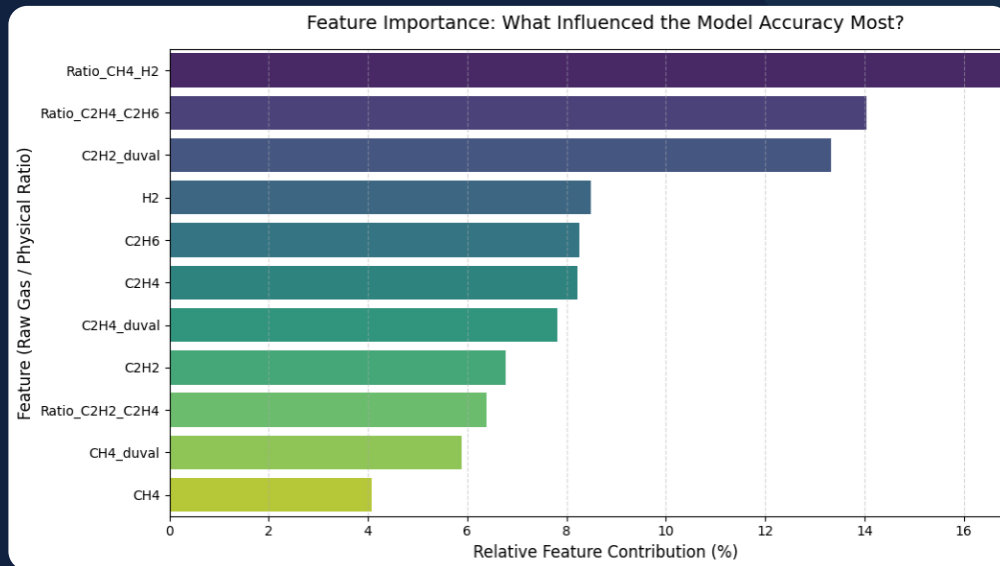


What Matters to the Model? (Feature Importance)

Value of Feature Engineering

Note that GridSearchCV selected the configuration **add_duval_param=True** and **add_rogers_param=True**.

- **Superiority of Physics:** Our generated features (Ratio, Duval) outperformed raw gas values in importance.
- **Leaders:** Gas ratios CH₄_H₂ and C₂H₄_C₂H₆ contribute the most to accuracy.
- **Conclusion:** Including industry expertise in the pipeline is critical for ML solution quality.



Conclusion and Key Takeaways

Key Project Results

1. **Efficiency:** Implementation of the CatBoost model achieved a classification accuracy of 92%, doubling the baseline performance.
2. **Expertise:** Feature Engineering based on physico-chemical indicators (Duval, Rogers methods) became the decisive factor for model quality.
3. **Reliability:** The model successfully handles class imbalance, ensuring high recognition accuracy for rare but critical defects.
4. **Outlook:** The developed pipeline is ready for integration into monitoring systems to prevent emergency situations.

Project Materials and Useful Links

Source Data and Code

DataSet: transformer-fault-diagnosis-dga-dataset
Open industrial dataset for 7 states.

GitHub: transformer-dga-ml
Code, EDA, transformers and pipeline .pkl.

Scientific and Methodological Base

IEC 60599:
Guide to the interpretation of dissolved and free gases analysis.

IEEE C57.104:
International standard for the interpretation of gases in oil-immersed transformers.

Professional Contacts

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